

Consumer Theory II: Utility and Demand

1 Consumer optimization

1. The general consumer optimization problem:

$$\begin{aligned} \max_{\{\mathbf{x} \in \mathbb{R}_+^n\}} u(\mathbf{x}) \\ s.t. \quad \mathbf{p}\mathbf{x} \leq I \end{aligned}$$

where $\mathbf{x} \in \mathbb{R}_+^n$ represents a consumption bundle, and I the income of the consumer.

2. Since the budget set, $B = \{\mathbf{x} \mid \mathbf{x} \in \mathbb{R}_+^n, \mathbf{p}\mathbf{x} \leq I\}$, is compact (closed and bounded) and u is continuous, by the Weierstrass theorem, there **exists** a solution to this problem. Moreover, if the budget set is convex and the objective function is strictly quasiconcave, the solution is **unique**. We call the solution to this problem

$$\mathbf{x}^* = \mathbf{x}(\mathbf{p}, I) \equiv (x_1(\mathbf{p}, I), \dots, x_n(\mathbf{p}, I))$$

the Marshallian demand function.

3. Weierstrass Theorem (Theorem A1.10): Let $f : S \rightarrow \mathbb{R}$ be a continuous real-valued function where S is a nonempty compact subset of $\mathbb{R}^n$. Then there exists a vector $x^* \in S$ and a vector $\tilde{x} \in S$ such that

$$f(\tilde{x}) \leq f(x) \leq f(x^*) \text{ for all } x \in S.$$

4. The Marshallian demand of good i , $x_i(p, I)$, is the quantity of good i the consumer would consume given the income and the prices. Note that the demand of a good depends on all prices. If we plot $x_i(p, I)$ against p_i , holding I and all prices other than p_i constant,

we get the demand curve of good i . A change in I or some p_j , $j \neq i$, would be represented by a shift of the demand curve.

5. Here are some basic properties of Marshallian demands:

- (a) $x(p, I)$ is homogeneous of degree 0 in p and I . That is, for any $t > 0$, $x(p, I) = x(tp, tI)$.
- (b) If u is strictly increasing, then $p \cdot x(p, I) = I$
- (c) If u is strictly quasiconcave, then $x(p, I)$ is unique.

6. Proof:

- (a) Multiplying both prices and income by the same factor leaves the budget set unchanged.
- (b) If u is increasing, the consumer can always increase his utility by consuming more of some goods.
- (c) Suppose consumption bundles x and x' , $x \neq x'$, both solve the consumer optimization problem. This necessarily means that $u(x) = u(x') = u^*$. Since u is strictly quasiconcave, it follows that

$$u(tx + (1 - t)x') > u^*,$$

a contradiction. Hence, the consumer optimization problem has a unique solution.

1.1 Differentiable Utility

1. We assume in this section that the utility function is twice differentiable.
2. Assuming differentiability brings two advantages. First, it allows to make precise statements about small changes. Second, It allows us to apply standard comparative statics techniques to examine how the optimal consumption bundle and welfare will change with prices and income.
3. The Two-Good Case

$$\begin{aligned} & \max_{x_1, x_2} u(x_1, x_2) \\ \text{s.t. } & p_1 x_1 + p_2 x_2 \leq I, \\ & x_1 \geq 0, \quad x_2 \geq 0. \end{aligned}$$

4. The Kuhn-Tucker Lagrangian is:

$$\tilde{L}(x_1, x_2, \lambda) = u(x_1, x_2) + \lambda(I - p_1x_1 - p_2x_2),$$

and the Kuhn-Tucker conditions are:

$$\begin{aligned} x_1 &\geq 0, \quad x_2 \geq 0 \\ \frac{\partial u}{\partial x_1} - \lambda p_1 &\leq 0, \quad \frac{\partial u}{\partial x_2} - \lambda p_2 \leq 0, \\ x_1(\frac{\partial u}{\partial x_1} - \lambda p_1) &= 0, \quad x_2(\frac{\partial u}{\partial x_2} - \lambda p_2) = 0, \\ \lambda &\geq 0, \quad \frac{\partial \tilde{L}}{\partial \lambda} = I - p_1x_1 - p_2x_2 \geq 0, \\ \lambda \frac{\partial \tilde{L}}{\partial \lambda} &= \lambda(I - p_1x_1 - p_2x_2) = 0. \end{aligned}$$

5. Assume u is strictly increasing. A rational consumer will fully spend his income. We can, without loss of generality, replace the inequality sign with equality sign in the budget constraint. So, we have the following two cases.
6. Case 1. Interior solution ($x_1 > 0, x_2 > 0$). The indifference curve must be tangent to the budget line at the optimal point. Analytically, this means that at the optimal point, the marginal rate of substitution must equal the relative price:

$$\frac{\frac{\partial u}{\partial x_1}}{\frac{\partial u}{\partial x_2}} = \frac{p_1}{p_2}. \quad (1)$$

The terms of trade $\frac{p_1}{p_2}$ is how many units of good 2 you can get by one unit of good 1, and MRS is how many units of good 2 you are willing to give up for one unit of good 1. At the optimal bundle, they should be equal. We can also write the above condition as

$$\frac{\frac{\partial u}{\partial x_1}}{p_1} = \frac{\frac{\partial u}{\partial x_2}}{p_2}. \quad (2)$$

7. Case 2. Corner solutions ($x_1 = 0$ or $x_2 = 0$). A necessary condition of a corner solution to be optimal is that the marginal utility per dollar for the good that the consumer is not consuming must be less than the marginal utility per dollar for the good that he is consuming. That is, $x_1 = 0, x_2 = I/p_2$ is optimal only if

$$\frac{\frac{\partial u}{\partial x_1}}{p_1} \leq \frac{\frac{\partial u}{\partial x_2}}{p_2}. \quad (3)$$

Or $x_2 = 0$, $x_1 = I/p_1$ is optimal only if

$$\frac{\frac{\partial u}{\partial x_1}}{p_1} \geq \frac{\frac{\partial u}{\partial x_2}}{p_2}. \quad (4)$$

Note that conditions (2), (3), and (4) are necessary but not sufficient for optimality. If more than one of three are held, then you need solve out the utility for each case and see which one is actually the highest.

8. The n -good Case. Returning to the general problem:

$$\begin{aligned} & \max u(x) \\ & s.t. \quad p \cdot x \leq I \\ & \quad x \geq 0 \end{aligned}$$

Let

$$\mathcal{L}(x, \lambda) \equiv u(x) + \lambda(I - px) \quad (5)$$

be the Lagrangian function. A consumption bundle $x^* \geq 0$ satisfies the first-order condition of optimization if there exists λ^* such that

$$\frac{\frac{\partial u(x^*)}{\partial x_i}}{p_i} = \lambda^* \quad \forall x_i^* > 0 \quad (6)$$

and

$$\frac{\frac{\partial u(x^*)}{\partial x_i}}{p_i} \leq \lambda^* \quad \forall x_i^* = 0. \quad (7)$$

The parameter λ^* measure the marginal utility of income. It is the extra utility an extra unit of income can buy.

9. The first-order condition is usually necessary condition, and we need add the second-order condition to form the sufficient conditions. But under certain assumption, the first first-order condition can be sufficient. For example: a sufficient condition for constrained maximization: When u in the maximization problem (5) is quasiconcave and strictly increasing, then the first-order condition (6) (7) is sufficient for optimality.

1.2 An Example: CES utility function

Suppose $u(x_1, x_2) = (x_1^\rho + x_2^\rho)^{1/\rho}$, where $0 \neq \rho < 1$. The utility function is strictly increasing and strictly quasiconcave.

The consumer's problem:

$$\max_{x_1, x_2} (x_1^\rho + x_2^\rho)^{1/\rho}$$

$$\text{subject to: } p_1x_1 + p_2x_2 \leq I, \quad x_1, x_2 \geq 0$$

To solve this problem, first write down the Lagrangian. Because the utility function is strictly increasing, the budget constraint will hold with equality. Assuming **the interior solution**, the **Kuhn-Tucker conditions** coincide with the ordinary first-order condition, therefore,

$$L(x_1, x_2, \lambda) = (x_1^\rho + x_2^\rho)^{1/\rho} + \lambda(I - p_1x_1 - p_2x_2)$$

By the first-order conditions:

$$\frac{\partial L}{\partial x_1} = (x_1^\rho + x_2^\rho)^{(1/\rho)-1} x_1^{\rho-1} - \lambda p_1 = 0$$

$$\frac{\partial L}{\partial x_2} = (x_1^\rho + x_2^\rho)^{(1/\rho)-1} x_2^{\rho-1} - \lambda p_2 = 0$$

$$\frac{\partial L}{\partial \lambda} = I - p_1x_1 - p_2x_2 = 0$$

1.2.1 Marshallian demands

By the first two equations, we can get

$$x_1 = x_2 \left(\frac{p_1}{p_2} \right)^{1/(\rho-1)}$$

By the third one, we have

$$I = p_1x_1 + p_2x_2$$

By the above two equations, we can get the Marshallian demands

$$x_1(p, I) = \frac{p_1^{r-1} I}{p_1^r + p_2^r}, \quad x_2(p, I) = \frac{p_2^{r-1} I}{p_1^r + p_2^r}$$

where $r = \rho/(\rho - 1)$.

1.2.2 Indirect utility function

We can also obtain the indirect utility function:

$$v(p, I) \equiv u(x(p, I)) = [(x_1(p, I))^\rho + (x_2(p, I))^\rho]^{1/\rho} = I(p_1^r + p_2^r)^{-1/r}.$$

1.3 Indirect Utility Function

1. The indirect utility function is the maximum utility a consumer can achieve given income and prices.
2. Definition: Indirect utility function:

$$v(p, I) \equiv u(x(p, I)).$$

3. It is the maximum-value function corresponding to the consumer's utility maximization problem.
4. Theorem. If $u(x)$ is continuous and strictly increasing on $\mathbb{R}_+^n$, then the indirect utility function $v(p, I)$ is

- (a) continuous on $\mathbb{R}_{++}^n \times \mathbb{R}_+$,
- (b) homogeneous of degree zero in (p, I) ,
- (c) strictly increasing in I ,
- (d) decreasing in p , and
- (e) quasiconvex in (p, I)
- (f) Roy's identity: If $v(p, I)$ is differentiable at (p^0, I^0) and $\partial v(p^0, I^0)/\partial I \neq 0$, then

$$x_i(p^0, I^0) = -\frac{\partial v(p^0, I^0)/\partial p_i}{\partial v(p^0, I^0)/\partial I}.$$

Property (e) says that a consumer would prefer one of any two extreme budget sets to any average of the two. Property (f) says that the consumer's Marshallian demand for good i is simply the ratio of the partial derivatives of indirect utility with respect to p_i and I after a sign change.

1. Proof:

- (a) Skip.
- (b) Multiplying both the prices and income by the same factor leaves the budget set unchanged. Hence, the maximum utility achievable is unchanged.
- (c) Let $I' > I$ and $e \equiv (1, \dots, 1)$

$$\begin{aligned} v(p, I) \equiv u(x(p, I)) &< u\left(x(p, I) + \frac{I' - I}{\sum_{i=1}^n p_i} e\right) \quad (u \text{ is strictly increasing}) \\ &< v(p, I') \text{ by definition of } v. \end{aligned}$$

- (d) Suppose $p^0 \geq p^1$. Then x^0 , the optimal consumption bundle at p^0 , is feasible at p^1 . Hence,

$$\begin{aligned} v(p^1, I) &= \max_x u(x) \text{ s.t. } p^1 x \leq I \\ &\geq u(x^0) \\ &= v(p^0, I). \end{aligned}$$

- (e) We need to show that for all $t \in [0, 1]$ and for all $(p, I), (p', I') \in \mathbb{R}_{++}^n \times \mathbb{R}_+$,

$$v(tp + (1-t)p', tI + (1-t)I') \leq \max\{v(p, I), v(p', I')\}.$$

Let

$$\begin{aligned} B_1 &\equiv \{x : px \leq I\} \\ B_2 &\equiv \{x : p'x \leq I'\} \\ B_3 &\equiv \{x : (tp + (1-t)p')x \leq tI + (1-t)I'\} \end{aligned}$$

Note that if $px > I$ and $p'x > I'$, then

$$(tp + (1-t)p')x > tI + (1-t)I'.$$

Thus, if $x \notin B_1 \cup B_2$, then $x \notin B_3$. The optimal bundle under $t(p, I) + (1-t)(p', I')$ must be feasible under either (p, I) or (p', I') .

- (f) Roy's identity: Using the Lagrangian function, we can write

$$v(p, I) = u(x^*) + \lambda(I - px^*).$$

Now applying **the envelope theorem**, we have

$$\frac{\partial v}{\partial p_i} = -\lambda x_i$$

and

$$\frac{\partial v}{\partial I} = \lambda.$$

Hence, the result.

2 The Dual Problem

2.1 Expenditure Minimization

1. The expenditure minimization problem:

$$\begin{aligned} \min_x & p \cdot x \\ \text{s.t.} & \quad u(x) \geq u, \\ & \quad x \geq 0. \end{aligned}$$

The solution to this problem tells us the minimum expenditure that is necessary such that the consumer's utility is not lower than u .

2. Solving this problem tells the decision-maker (e.g. the government) the payment necessary to keep a person's utility above a minimum level.
3. Definitions: A solution to the expenditure minimization problem is known as the Hicksian demand function and is denoted by

$$x^h(p, u).$$

Substituting $x^h(p, u)$ into the objective function gives us the expenditure function

$$e(p, u) \equiv p \cdot x^h(p, u).$$

4. Properties of Hicksian Demand: Suppose u is continuous and strictly increasing, then x^h has the following properties:
 - (a) Homogeneity of degree zero in p . That is, $x^h(tp, u) = x^h(p, u)$ for all scalar $t > 0$.
 - (b) No excess utility: $u(x^h(p, u)) = u$.
5. These properties are similar to those of the Marshallian demand. Briefly, (a) is obvious from the defin. of the minimization problem; (b) follows from strict monotonicity.
6. Theorem: Suppose u is strictly increasing and $x^h(p, u)$ is unique for all p . Then for any p' and p'' ,

$$(p'' - p') \cdot (x^h(p'', u) - x^h(p', u)) \leq 0.$$

7. Proof. We note that

$$u(x^h(p'', u)) = u(x^h(p', u)).$$

By definition, $x^h(p'', u)$ and $x^h(p', u)$ are the cheapest way to achieve u under p'' and p' , respectively. Hence

$$\begin{aligned} p'' \cdot x^h(p'', u) &\leq p'' \cdot x^h(p', u), \\ p' \cdot x^h(p', u) &\leq p' \cdot x^h(p'', u). \end{aligned}$$

These two inequalities imply that

$$(p'' - p') \cdot (x^h(p'', u) - x^h(p', u)) \leq 0.$$

Note that this result follows entirely from the fact that x^h minimizes expenditure. It does not use any other properties of the utility functions.

8. This result is called the compensated law of demand. It states that an expenditure-minimizing consumer should substitute low-price goods for high-price goods. Note that if p'' and p' differ only in the price of good i , then the result becomes

$$(p''_i - p'_i) (x^h_i(p'', u) - x^h_i(p', u)) \leq 0,$$

which means that the Hicksian demand for good i is downward sloping.

9. Properties of the Expenditure Function: If $u(\cdot)$ is continuous and strictly increasing, then the expenditure function $e(p, u)$ has the following properties:

- (a) Zero when u takes on the lowest level of utility (i.e $u(0)$ when the utility function is increasing).
- (b) Continuous on its domain $\mathbb{R}_{++}^n \times U$.
- (c) For all $p \gg 0$, strictly increasing and unbounded above in u .
- (d) Increasing in p .
- (e) Homogeneous of degree 1 in p .
- (f) Concave in p .

10. Proof:

- (a) The consumer can achieve $u(0)$ by not spending anything.

- (b) Intuitively, the result follows from the continuity of u and the continuity of the budget set.
- (c) This is straightforward. Suppose e is not strictly increasing in u . Then there exist p, u_0 and $u_1, u_1 > u_0$, such that $e(p, u_1) \leq e(p, u_0)$. Since u is continuous and strictly increasing in x , it is possible to reduce consumption by some small ε such that

$$u(x^h(p, u_1) - \varepsilon) > u_0$$

and

$$p \cdot (x^h(p, u_1) - \varepsilon) < e(p, u_0).$$

Here ε is a vector. This violates the definition of $e(p, u_0)$. Suppose that e is bounded by some e^* (meaning that $e(p, u) \leq e^*$ for all p and u). Since e is strictly increasing in u , $e(p, v(p, e^*) + 1)$ must be greater than e^* , a contradiction.

- (d) If $p^1 \geq p^0$ (that is, $p_i^1 \geq p_i^0$ for all i), then any consumption bundle x that is feasible under p^1 is also under p^0 . Since $e(p, u)$ is by definition the lowest expenditure that is necessary to achieve u ,

$$e(p^0, u) \leq p^0 x^h(p^1, u) \leq p^1 x^h(p^1, u) \equiv e(p^1, u).$$

- (e) This immediately follows from the fact the Hicksian demand is homogeneous in degree zero in p .
- (f) Let $x^1 = x^h(p^1, u)$ and $x^2 = x^h(p^2, u)$. By definition, for any x such that $u(x) \geq u$

$$e(p^1, u) = p^1 x^1 \leq p^1 x$$

and

$$e(p^2, u) = p^2 x^2 \leq p^2 x.$$

It follows that for any $t \in [0, 1]$ and for any x such that $u(x) \geq u$

$$te(p^1, u) + (1-t)e(p^2, u) \leq (tp^1 + (1-t)p^2)x.$$

The above equation holds for all x such that $u(x) \geq u$, including $x^h(tp^1 + (1-t)p^2, u)$. Hence,

$$\begin{aligned} te(p^1, u) + (1-t)e(p^2, u) &\leq (tp^1 + (1-t)p^2)x^h(tp^1 + (1-t)p^2, u) \\ &\equiv e((tp^1 + (1-t)p^2), u). \end{aligned}$$

It is important to note that the concavity of e follows from the definition of e and does not require that u be quasiconcave.

11. Differentiable Utility. Solving a minimization problem involves the same technique as solving a maximization problem. However, in this problem the last equation comes from the utility constraint, whereas in the maximization problem it comes from the budget constraint. The Lagrangian function is

$$\mathcal{L} = p \cdot x + \lambda(u - u(x)).$$

The first order conditions are (assuming $x \gg 0$):

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x_i} &= p_i - \lambda \frac{\partial u}{\partial x_i} = 0 \text{ for all } i = 1, \dots, n, \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= u - u(x) = 0. \end{aligned}$$

The first n equations imply

$$\frac{\frac{\partial u}{\partial x_1}}{p_1} = \dots = \frac{\frac{\partial u}{\partial x_n}}{p_n}$$

12. Shephard's lemma: If $e(p, u)$ is differentiable in p at (p^0, u^0) with $p^0 \gg 0$, then

$$\frac{\partial e(p^0, u^0)}{\partial p_i} = x_i^h(p^0, u^0), \quad i = 1, \dots, n.$$

The results follows directly from the envelope theorem.

$$\frac{\partial e}{\partial p_i} = \frac{\partial \mathcal{L}}{\partial p_i} = x_i^h.$$

The Shephard's lemma is the counterpart of Roy's Identity. Same Intuition.

2.2 Relationship Between Marshallian and Hicksian Demand Functions

1. The indirect utility function and expenditure function are intimately connected. So are the Marshallian and Hicksian demand functions. We have already seen that the two demand functions share a common tangency condition. Regardless of whether one's objective is to maximize utility or minimize expenditure, one should consume in such a way that the marginal utility per dollar spent on each good is the same.

2. Theorem 1.8. Let $v(p, I)$ and $e(p, u)$ be the indirect utility function and expenditure function for a consumer with a continuous and strictly increasing utility function. Then

$$(a) \quad e(p, v(p, I)) = I,$$

$$(b) \quad v(p, e(p, u)) = u.$$

3. Part (a) says that if the maximum utility the consumer can obtain given p and I is v , then the minimum expenditure needed to achieve v is I . Part (b) says that if the minimum expenditure needed to achieve utility u given p is e , then maximum utility one can obtain given p and e is u .

4. Proof: I will only prove part (a). Part (b) can be proven in the same way. Since $e(p, v(p, I))$ is by definition the minimum expenditure needed to attain $v(p, I)$, it must be that $e(p, v(p, I)) \leq I$.

Suppose $e(p, v(p, I)) < I$. Then, there exists x^* such that $px^* < I$ and $u(x^*) \geq v(p, I)$. By strict monotonicity, $p(x^* + \varepsilon) < I$ and $u(x^* + \varepsilon) > v(p, I)$ for some small ε . This contradicts the supposition that $v(p, I)$ is the highest utility attainable given p and I .

5. Note that the theorem is incorrect if the utility function is not strictly increasing. For example: if $u(x) = 10$ for all x , then

$$v(p, 5) = 10,$$

$$e(p, 10) = 0.$$

6. Theorem 1.9. Suppose u is continuous, strictly quasiconcave, and strictly increasing. Then for all $p \gg 0$, $I \geq 0$, and $u \in U$, we have

$$(a) \quad x(p, I) = x^h(p, v(p, I)).$$

$$(b) \quad x^h(p, u) = x(p, e(p, u)).$$

7. The first relation says that the Marshallian demand at prices p and income I is equal to the Hicksian demand at prices p and the maximum utility level that can be achieved at prices p and income I . The second says that the Hicksian demand at any prices p and utility level u is the the same as the Marshallian demand at those prices and an income level equal to the minimum expenditure necessary to achieve the utility level.

8. Proof: Strict quasiconcavity means that both $x(p, I)$ and $x^h(p, u)$ are unique. To prove part (a), note that by Theorem 1.8 the minimum expenditure needed to attain $v(p, I)$ is I . Since $u(x(p, I)) = v(p, I)$ (by definition of v) and $px(p, I) = I$, $x(p, I)$ is the solution of the expenditure minimization problem. The proof for the second part is similar.
9. There are two ways to derive the Hicksian demand. The direct way is to solve expenditure minimization problem. But we can also get the marshallian demand by solving the utility maximization problem. Substitute the demand back into the utility function. This gives the indirect utility function. From the indirect utility function, we know how much income need to change to keep utility constant as price change. Substitute this back into the Marshallian demand gives the Hicksian demand.
10. Cobb-Douglas example: $u(x_1, x_2) = x_1^\alpha x_2^\beta$, $\alpha + \beta = 1$

$$x_1 = \frac{\alpha I}{p_1}, x_2 = \frac{\beta I}{p_2}$$

The indirect utility function is

$$v(I, p_1, p_2) = u(x(p_1, p_2, I), x_2(p_1, p_2, I)) = \left(\frac{\alpha I}{p_1}\right)^\alpha \left(\frac{\beta I}{p_2}\right)^\beta = \left(\frac{\alpha}{p_1}\right)^\alpha \left(\frac{\beta}{p_2}\right)^\beta I$$

Rearranging the terms

$$e(p_x, p_y, u) = \frac{u}{\left(\frac{\alpha}{p_1}\right)^\alpha \left(\frac{\beta}{p_2}\right)^\beta} = \left(\frac{p_1}{\alpha}\right)^\alpha \left(\frac{p_2}{\beta}\right)^\beta u.$$

The Hicksian demand for Cobb-Douglas utility function is

$$x_1^h = \left(\frac{p_1}{\alpha}\right)^{\alpha-1} \left(\frac{p_2}{\beta}\right)^\beta u, x_2^h = \left(\frac{p_1}{\alpha}\right)^\alpha \left(\frac{p_2}{\beta}\right)^{\beta-1} u.$$

2.3 An Example: CES (continued)

We continue the example $u(x_1, x_2) = (x_1^\rho + x_2^\rho)^{1/\rho}$. We want to derive the corresponding expenditure function, then compare it with the indirect utility function, and show the duality between Marshallian and Hicksian demand functions.

2.3.1 Expenditure function

Because the preferences are monotonic, we can formulate the expenditure minimization problem as:

$$\min_{x_1, x_2} p_1 x_1 + p_2 x_2$$

$$\text{subject to: } (x_1^\rho + x_2^\rho)^{1/\rho} = u, \quad x_1, x_2 \geq 0$$

And its Lagrangian

$$L(x_1, x_2, \lambda) = p_1 x_1 + p_2 x_2 + \lambda[u - (x_1^\rho + x_2^\rho)^{1/\rho}]$$

Assuming the interior solution in both goods, and by the first-order condition we have:

$$\frac{\partial L}{\partial x_1} = p_1 - \lambda(x_1^\rho + x_2^\rho)^{(1/\rho)-1} x_1^{\rho-1} = 0$$

$$\frac{\partial L}{\partial x_2} = p_2 - \lambda(x_1^\rho + x_2^\rho)^{(1/\rho)-1} x_2^{\rho-1} = 0$$

$$\frac{\partial L}{\partial \lambda} = u - (x_1^\rho + x_2^\rho)^{1/\rho} = 0$$

By the first two, we can get

$$x_1 = x_2 \left(\frac{p_1}{p_2} \right)^{1/(\rho-1)}$$

By the third one, we have

$$u = (x_1^\rho + x_2^\rho)^{1/\rho}$$

2.3.2 Hicksian demands

By the above two equations, we can get the Hicksian demands

$$x_1^h(p, u) = u(p_1^r + p_2^r)^{(1/r)-1} p_1^{r-1}, \quad x_2^h(p, u) = u(p_1^r + p_2^r)^{(1/r)-1} p_2^{r-1}$$

where $r = \rho/(\rho - 1)$.

2.3.3 Expenditure function

Therefore, we can obtain the expenditure function:

$$e(p, u) = p_1 x_1^h(p, u) + p_2 x_2^h(p, u) = u(p_1^r + p_2^r)^{1/r}.$$

2.3.4 Relationship between indirect utility and expenditure functions

They actually imply each other. When you get one, you can get the other very quickly. In previous lecture, we've already got the indirect utility function:

$$v(p, I) = I(p_1^r + p_2^r)^{-1/r}.$$

For income level $I = e(p, u)$, we must have

$$v(p, e(p, u)) = e(p, u)(p_1^r + p_2^r)^{-1/r}$$

We've already know from theorem 1.8, that $v(p, e(p, u)) = u$, therefore we have

$$u = e(p, u)(p_1^r + p_2^r)^{-1/r}$$

We can get the expenditure function

$$e(p, u) = u(p_1^r + p_2^r)^{1/r}$$

And you can try to get the indirect utility function given the expenditure function is known.

2.3.5 Relationship between Marshallian and Hicksian demands

We know the Marshallian demands in the previous lecture as

$$x_i(p, I) = \frac{p_i^{r-1} I}{p_1^r + p_2^r}, \quad i = 1, 2,$$

$$\begin{aligned} x_i(p, e(p, u)) &= \frac{e(p, u) p_i^{r-1}}{p_1^r + p_2^r} \\ &= u(p_1^r + p_2^r)^{(1/r)-1} p_i^{r-1} \\ &= x_i^h(p, u) \end{aligned}$$

3 Other Topics

3.1 Income and Substitution Effects

1. Slutsky equation: since

$$x_i^h(p, u^*) = x_i(p, e(p, u^*))$$

$$\begin{aligned}
\frac{\partial x_i^h(p, u^*)}{\partial p_j} &= \frac{\partial x_i(p, I)}{\partial p_j} + \frac{\partial x_i(p, I)}{\partial I} \frac{\partial e(p, u^*)}{\partial p_j} \\
&= \frac{\partial x_i(p, I)}{\partial p_j} + \frac{\partial x_i(p, I)}{\partial I} x_j^h(p, u^*) \quad (\text{by Sheppard's lemma}) \\
&= \frac{\partial x_i(p, I)}{\partial p_j} + x_j(p, I) \frac{\partial x_i(p, I)}{\partial I}
\end{aligned}$$

- Rearranging the terms, we have

$$\frac{\partial x_i(p, I)}{\partial p_j} = \frac{\partial x_i^h(p, u^*)}{\partial p_j} - x_j(p, I) \frac{\partial x_i(p, I)}{\partial I} \quad (8)$$

The Slutsky equation decompose the effect of the price of good j on the quantity demand of good i into a substitution effect and an income effect. The former captures the effect of relative price, and the latter the effect of purchasing power.

- To separate the income effect from the substitution effect, we can imagine that the consumers are compensated for the change in purchasing power due to a change in price so that their utility remain unchanged, and study how their demand for a good will change purely as a result of relative price changes. Hence, the Hicksian demand is also called the compensated demand, and the Marshallian Demand uncompensated demand.
- From the theorem, we certainly have

$$\frac{\partial x_i(p, I)}{\partial p_i} = \frac{\partial x_i^h(p, u^*)}{\partial p_i} - x_i(p, I) \frac{\partial x_i(p, I)}{\partial I}$$

- $\frac{\partial x_i^h(p, u^*)}{\partial p_i} \leq 0$. The reason is because the expenditure function is a concave function of p , therefore we have

$$\frac{\partial^2 e(p, u)}{\partial p_i^2} = \frac{\partial x_i^h(p, u)}{\partial p_i} \leq 0$$

The income effect can be positive or negative. Thus, (8) tells that for normal goods ($\frac{\partial x_i}{\partial I} > 0$),

$$\frac{\partial x_i}{\partial p_j} < \frac{\partial x_i^h}{\partial p_j}$$

meaning that the Marshallian demand has a flatter slope. The opposite is true for inferior goods.

3.2 The Slutsky Substitution Matrix

1. Given a Marshallian demand function $x(p, I)$, let

$$s_{ij} \equiv \frac{\partial x_i(p, I)}{\partial p_j} + x_j(p, I) \frac{\partial x_i(p, I)}{\partial I}.$$

2. The Slutsky matrix is defined as

$$\begin{bmatrix} s_{11} & s_{12} & \dots & s_{1n} \\ s_{21} & & & \\ \cdot & & \cdot & \\ \cdot & & \cdot & \\ s_{n1} & & & s_{nn} \end{bmatrix}$$

3. Theorem: Suppose $x(p, I)$ is a Marshallian demand function generated by some continuous, strictly increasing utility function. Then the Slutsky matrix of x is symmetric and negative semidefinite.
4. Proof: The proof is not hard, but it makes use of several results that we have just learnt. From the Slutsky equation, we have

$$s_{ij} = \frac{\partial x_i^h}{\partial p_j}.$$

(The reason we need the utility function to be strictly increasing or at least locally non-satiated, is that without it Theorem 1.9 and, hence, the Slutsky equation does not hold.) By the Sheppard's lemma, we have for any i and j

$$\frac{\partial^2 e}{\partial p_i \partial p_j} = \frac{\partial x_i^h}{\partial p_j} = \frac{\partial x_j^h}{\partial p_i}.$$

This implies that the Slutsky matrix is symmetric. Finally, since e is concave, its Hessian matrix:

$$\begin{bmatrix} \frac{\partial x_1^h}{\partial p_1} & \frac{\partial x_1^h}{\partial p_2} & \dots & \frac{\partial x_1^h}{\partial p_n} \\ \frac{\partial x_2^h}{\partial p_1} & & & \\ \cdot & & \cdot & \\ \cdot & & \cdot & \\ \frac{\partial x_n^h}{\partial p_1} & & & \frac{\partial x_n^h}{\partial p_n} \end{bmatrix}$$

must be negative semidefinite.

3.3 Integrability

1. The problem of recovering a consumer's utility function from his or her demand function is known as the integrability problem.
2. Demand behavior is consistent with the theory of utility maximization if and only if it satisfies budget balanceness, negative semidefiniteness, and symmetry. Formally,
3. Integrability Theorem (Theorem 2.6, J&R). A continuous differentiable function $\mathbf{x}: \mathbf{R}_{++}^{n+1} \rightarrow \mathbf{R}_+^n$ is the demand function generated by some increasing, quasiconcave utility function if (and only if, when utility is continuous, strictly increasing, and strictly quasiconcave) it satisfies budget balanceness, symmetry, and negative semidefiniteness. (See J&R page 88-90 for the proof)

3.4 Revealed Preference

1. We have begun the consumer theory by assuming things we cannot observe (preference) to make predictions about something we can observe (consumer demand behavior). Why not start with the observable behavior?
2. Samuelson (1947) shows that every prediction consumer theory makes can also be derived from a few simple and sensible assumptions about the consumer's observable choices. The idea is simple: If the consumer buys one bundle instead of another affordable bundle, then the first bundle is considered to be **revealed preferred** to the second. We thus make assumptions about the consistency of the choices that are made:
3. **Weak Axiom of Revealed Preference (WARP)**. A consumer's choice behavior satisfies WARP if for every distinct pair of bundles $\mathbf{x}^0, \mathbf{x}^1$ with $\mathbf{x}^0$ chosen at prices p^0 and $\mathbf{x}^1$ chosen at prices p^1 ,

$$p^0 \mathbf{x}^1 \leq p^0 \mathbf{x}^0 \Rightarrow p^1 \mathbf{x}^0 > p^1 \mathbf{x}^1.$$

In other words, WARP holds if whenever $\mathbf{x}^0$ is revealed preferred to $\mathbf{x}^1$, $\mathbf{x}^1$ is never revealed preferred to $\mathbf{x}^0$.

4. A choice function $\mathbf{x}(\mathbf{p}, I)$ satisfies WARP and budget balance, then it must satisfy homogeneity of degree zero and negative semidefiniteness of the Slutsky matrix. With two-good case, budget balanceness together with homogeneity imply the Slutsky matrix must be symmetric. Therefore, for two goods, the integrability theorem tells us the choice function must be utility-generated demand function.

5. For more than two goods, we need **Strong Axiom of Revealed Preference (SARP)**: If for every sequence of distinct bundles $\mathbf{x}^0, \mathbf{x}^1, \dots, \mathbf{x}^k$, where $\mathbf{x}^0$ is revealed preferred to $\mathbf{x}^1$, and $\mathbf{x}^1$ is revealed preferred to $\mathbf{x}^2, \dots$, and $\mathbf{x}^{k-1}$ is revealed preferred to $\mathbf{x}^k$, it is not the case that $\mathbf{x}^k$ is revealed preferred to $\mathbf{x}^0$. SARP rules out intransitive revealed preferences and thus can be used to induce a complete and transitive preference relation $\succsim$, for which there will then exist a utility function that **rationalises** the observed behavior.

3.5 Compensating Variation and Consumer Surplus

1. When the government decides whether to cut or raise sales taxes, it has to know how the tax affects consumer welfare. In antitrust law suit, an important factor that the court will look into is whether consumers are hurt by the monopolist.
2. The expenditure function provides us a formal way to measure how a change in price affects consumer welfare. Definition: Compensating variation (CV) associated with a change in the price of good i is the income compensation required to keep the utility of a consumer constant. That is,

$$CV \equiv e(p'_i, p_{-i}, u) - e(p_i, p_{-i}, u).$$

Or, alternatively,

$$v(p'_i, p_{-i}, I + CV) = v(p_i, p_{-i}, I).$$

By Shephard's Lemma, $\frac{\partial e}{\partial p_i} = x^h$. It follows that

$$\begin{aligned} CV &= \int_{p_i}^{p'_i} \frac{\partial e}{\partial p_i} dp_i, \\ &= \int_{p_i}^{p'_i} x^h(p_i, p_{-i}, u) dp_i. \end{aligned}$$

Thus, the compensating variation is the area to the left of the Hicksian demand between p_i and p'_i . (Draw diagram) This is an extremely important result because it allow us to evaluate the welfare consequence of a policy.

3. Since Hicksian demand is not directly observable, it is more convenient to measure welfare change using Marshallian rather than Hicksian demand. The consumer surplus (at p_i), CS, is the area

bound by the Marshallian demand and p_i . The change in CS when the price of good i changes from p_i to p'_i is

$$\Delta CS = \int_{p'_i}^{p_i} x(p_i, p_{-i}, I) dp_i.$$

Recall from the Slutsky equation that

$$\left. \frac{\partial x_i}{\partial p_j} \right|_{p,I} = \left. \frac{\partial x_i^h}{\partial p_j} \right|_{p,v(p,I)} - \left. \frac{\partial x_i}{\partial I} \right|_{p,I} x_i(p, I).$$

For normal goods, $x_i(p'_i, I) < x_i^h(p'_i, v(p_i, p_{-i}, I))$ when $p'_i > p_i$, that is, Marshallian demand decreases more when it is a price increase. $x_i(p'_i, I) > x_i^h(p'_i, v(p_i, p_{-i}, I))$ when $p'_i < p_i$, that is, Marshallian demand increases more when price goes down. The converse is true for inferior goods. Hence, for normal goods

$$|\Delta CS| > |CV|$$

if price decreases, and

$$|\Delta CS| < |CV|$$

when price increases. For normal goods, the change in consumer surplus underestimates the welfare loss due to price increase and overestimate the welfare gain due to a price decrease.

4. Although, $|\Delta CS| \neq |CV|$, in practice the difference is pretty small and is insignificant compared to measurement errors.